

Job week - 1

24/4/6

Job sequence

Consider a set of 14 jobs, where each job require one unit of time for completion. Each job has an associated profit and deadline.

↓ 22, 19, 29, 28, 32, 27, 25, 24, 26, 14, 27, 19, 119

deadlines of the jobs are :-

↓ 2, 3, 8, 6, 7, 5, 10, 4, 6, 12, 13, 2, 10, 14

profit :- { 80, 29, 28, 27, 27, 26, 20, 24, 22, 21, 19, 19, 14

Deadlines :- { 7, 8, 6, 10, 2, 12, 14, 6, 3, 5, 3, 14, 13, 14

Jobs :- { J₁, J₂, J₃, J₄, J₅, J₆, J₇, J₈, J₉, J₁₀, J₁₁, J₁₂, J₁₃, J₁₄

J₁₁, J₁₄

Time slots

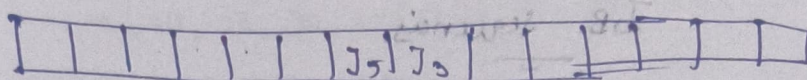
1) Add J₅ (30, 7)

J₅

profit :- 30

2)

$$J_3(29, 8)$$

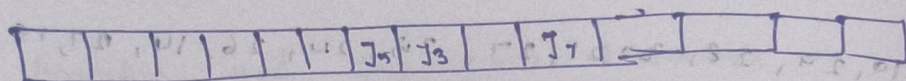


$$S = \{J_5, J_3\}$$

$$p = 59$$

3)

$$J_7(27, 10)$$

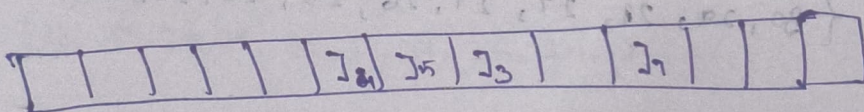


$$S = \{J_5, J_3, J_7\}$$

$$p = 114$$

4)

$$J_4(28, 6)$$

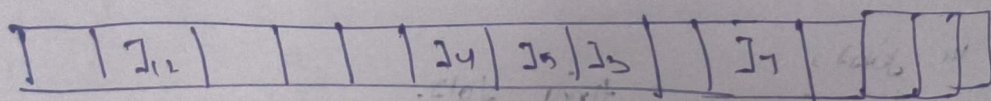


$$S = \{J_5, J_3, J_4\}$$

$$p = 87$$

5)

$$\text{Add } J_{12}(27, 2)$$

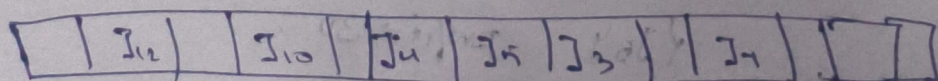


$$S = \{J_5, J_3, J_4, J_7, J_{12}\}$$

$$p = 141$$

6)

$$J_{10}(26, 4)$$



$$S = \{J_5, J_3, J_4, J_1, J_{12}, J_{10}\}$$

CH-26.04.25
24/4/25

$$P = \{1674\}$$

1) add $J_8 (25, 4)$

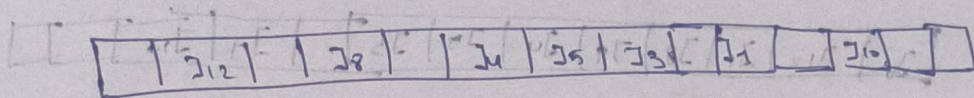
3 (1) 1674

1674

1674

1674

1674



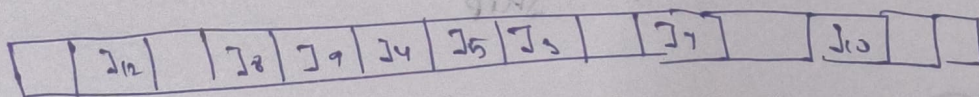
$$S = \{J_5, J_3, J_4, J_1, J_{12}, J_{10}, J_8\}$$

$$P = 192$$

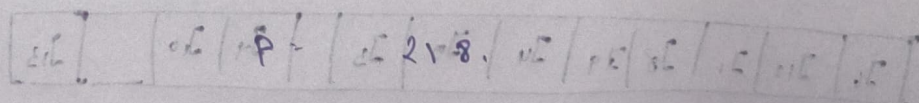
2)

$$J_9 (24, 6)$$

J_9 cannot be added to
can be added to slot 5.



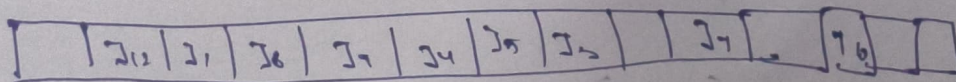
$$S = \{J_5, J_3, J_4, J_1, J_{12}, J_{10}, J_8, J_9\}$$



3)

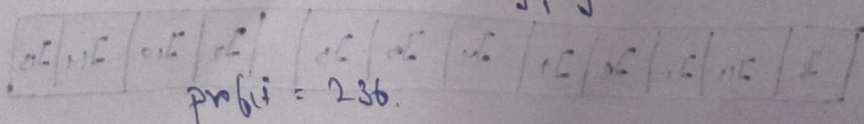
$J_{11} (22, 3)$ since empty at 3,

since empty at 3,



$$S = \{J_5, J_3, J_4, J_1, J_{12}, J_{10}, J_8, J_9\}$$

$$J_1, J_9$$



$$profit = 236$$

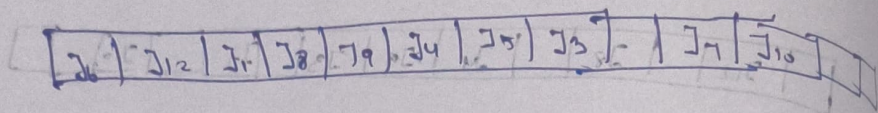
10) $J_6(21, 5)$ Not in 5

Connect in 5

4x
3x
2x
1x ✓

11/11/19

(11/11/19) 8x



$$S = \{ J_6, J_3, J_4, J_7, J_{12}, J_{10}, J_8, J_1, J_5, J_9 \}$$

p = 259.

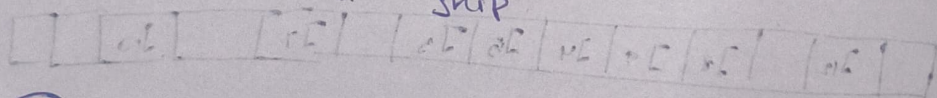
SP1 19

11) $J_2(19, 3)$

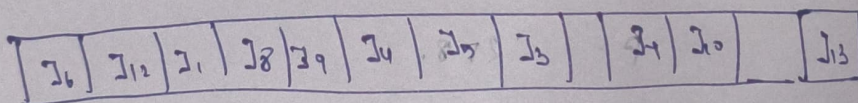
(11/11/19) 1x

Not in 3x
1x

Ship



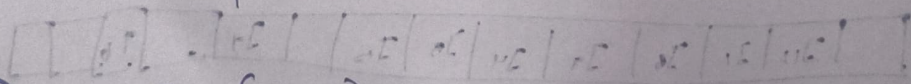
12) $J_{13}(19, 14)$



$$S = \{ J_6, J_3, J_4, J_7, J_{12}, J_{10}, J_8, J_1, J_{13}, J_5, J_9 \}$$

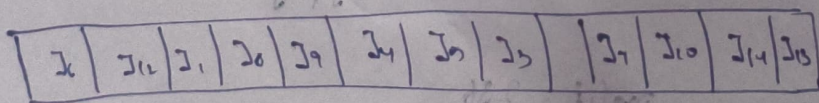
11/11/19

p = 278



13) $J_{11}(14, 13)$

11/11/19



$$S = \{ J_6, J_3, J_4, J_7, J_{12}, J_{10}, J_8, J_1, J_{13}, J_5, J_9 \}$$

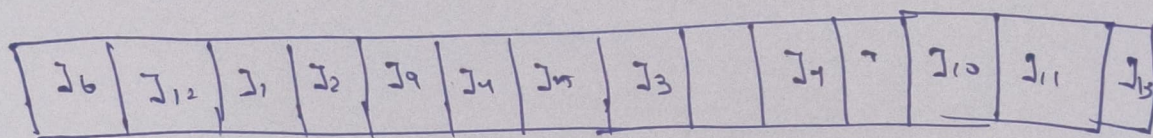
profit = 292

24146

n) $I_n(11,1)$

already occupies.

Find sequence



using greedy sequence, the optimal sequence of jobs

gives maximum profit of 293.